Snowflake AI for Commercial Analytics

Cortex Search · Cortex Analyst · Snowflake Intelligence

Pharma Commercial Team Demo | 2026

Vivitrol · Aristada · Lybalvi

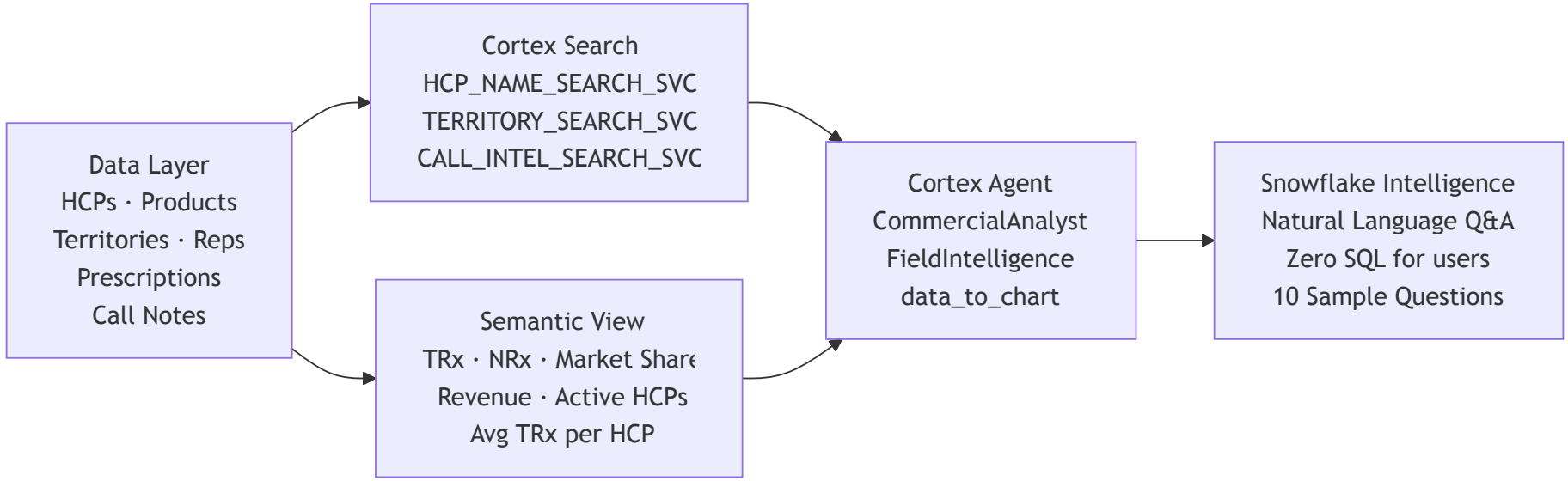
All data + AI in Snowflake — no external services, no data movement, zero SQL for end users.

The Commercial Analytics Challenge

- **Fragmented HCP data** — prescriber info lives in multiple systems; a single misspelling breaks a query
- **Rep call notes siloed in CRM** — qualitative field intelligence locked in free-text, invisible to BI tools
- **Formulary data goes stale** — payer barriers surface only in field reports, not dashboards
- **Business questions require the data team** — every insight becomes a SQL ticket with a waiting queue
- **No bridge between structured + unstructured data** — Rx metrics live in tables; physician objections live in text

> *"Which psychiatrists in Texas haven't had a rep call in 60 days?" should take seconds — not a ticket.*

Solution Architecture



One platform. All data + AI in Snowflake. No external services, no API keys, no data movement.

Data Model

Table	Contents	~Rows
`HCPS`	500 HCPs — name, NPI, specialty, city, state, tier	500
`PRESCRIPTIONS`	Monthly TRx & NRx by HCP, product, and territory	~12,000
`CALL_ACTIVITY`	Rep-HCP calls — type, outcome, products discussed	~15,000
`CALL_NOTES`	Rich free-text rep field notes — access issues, objections, feedback	30
`PRODUCTS`	Vivitrol, Aristada, Aristada Initio, Lybalvi catalog	4
`TERRITORIES`	50 US territories → district → region hierarchy	50
`SALES_REPS`	50 field reps with territory and district assignments	50
`MARKET_ACCESS`	Payer formulary status by product & plan (50 payers)	~600

Synthetic data modeled on CNS / addiction medicine specialties across all U.S. regions.

Cortex Search: Three Ways to Use It

Mode	What It Does	Key Demo Moment
1 — Analyst-Integrated	CS linked to a Semantic View dimension; fuzzy name resolution before SQL generation	"TRx for Dr. Andersen" → CS resolves to "Dr. Anderson" → SQL returns 42 TRx *(instead of 0)*
2 — Standalone	Direct `SEARCH_PREVIEW()` for semantic/fuzzy search on HCP profiles and call notes	"addiction psychiatry Texas" → ranked HCP list with scores, no exact match required
3 — Agent Tool	CS as a RAG retrieval tool inside the Cortex Agent, alongside Cortex Analyst	"Vivitrol formulary barriers" → agent retrieves top 5 matching rep call note excerpts

Each mode is independently valuable. Together they cover the full spectrum — structured analytics to pure unstructured retrieval — with the agent routing automatically between them.

Demo: Mode 1 — Cortex Search Inside Cortex Analyst

WITHOUT Cortex Search

```
-- User asks: "Show TRx for Dr. Andersen"
-- Analyst generates exact string match:

WHERE hcp_full_name = 'Dr. Andersen'
-- ↑ no exact match in database

-- Result: 0 rows returned ✗
```

Name misspelling → broken query → user thinks there's no data.

WITH Cortex Search on hcp_name dimension

```
-- CS resolves at query time:
--   "Dr. Andersen"
--     → "Dr. James Anderson"
-- Analyst generates correct SQL:

WHERE hcp_full_name = 'Dr. James Anderson'
-- ↑ fuzzy match resolved ✓

-- Result: 42 TRx returned ✓
```

Same question. Correct answer.

> ***How:*** `WITH CORTEX SEARCH SERVICE HCP_NAME_SEARCH_SVC` *in the Semantic View DDL links the service to the* `hcp_name` *dimension. Cortex Analyst calls it at query time to resolve literal values before generating SQL.*

Demo: Mode 2 — Standalone Cortex Search

Query HCP profile search service directly

```
SELECT PARSE_JSON(
  SNOWFLAKE.CORTEX.SEARCH_PREVIEW(
    'ALKERMES_DEMO', 'COMMERCIAL',
    'HCP_PROFILE_SEARCH_SVC',
    OBJECT_CONSTRUCT(
      'query', 'addiction medicine Texas',
      'columns', ARRAY_CONSTRUCT(
        'HCP_FULL_NAME', 'SPECIALTY', 'CITY'),
      'limit', 5
    )::VARCHAR
  )
) AS results;
```

No exact match required. Works with partial names, specialties, and location combinations.

Example results

HCP Name	Specialty	City
Dr. Sarah Martinez	Addiction Medicine	Dallas
Dr. James Brown	Psychiatry	Houston
Dr. Patricia Davis	Addiction Medicine	Austin
Dr. Thomas Garcia	Addiction Psychiatry	San Antonio

Key message: Snowflake-native semantic search — no Elasticsearch, no vector database, no custom embedding pipelines needed.

Semantic View: The Business Metric Layer

Metrics — defined once, queryable in plain English

Metric	Definition
`TOTAL_TRX`	Total prescriptions dispensed
`TOTAL_NRX`	New patient prescriptions written
`MARKET_SHARE_PCT`	Product TRx ÷ total market TRx × 100
`TOTAL_REVENUE_USD`	TRx × product list price
`ACTIVE_HCPS`	HCPs with ≥ 1 TRx in period
`AVG_TRX_PER_HCP`	TRx ÷ active HCP count

Dimensions — with synonyms & Cortex Search linkage

Dimension	Cortex Search
`HCP_NAME`	✅ `HCP_NAME_SEARCH_SVC`
`TERRITORY`	✅ `TERRITORY_SEARCH_SVC`
`PRODUCT`	sample values
`SPECIALTY`	sample values
`REGION`	sample values
`REP_NAME`	—
`RX_MONTH`	—

> *Define business logic once — no conflicting metric definitions across reports. Every Cortex Analyst question uses the same correct calculation.*

Demo: Cortex Agent Routing

The agent selects the right tool automatically — no user configuration required.

User Question	Tool Used	What Happens
"Which territories underperform on Vivitrol TRx this quarter?"	**CommercialAnalyst**	Text-to-SQL on semantic view → ranked territory table + bar chart
"Find field notes about physician objections to Lybalvi"	**FieldIntelligence**	Semantic search on `CALL_INTEL_SEARCH_SVC` → top 5 rep note excerpts with date and territory
"Which territories underperform AND what are reps saying?"	**Both tools**	Analyst returns metrics; Search retrieves qualitative context; combined into one answer

> *Routing instruction in agent spec: "Use CommercialAnalyst for quantitative questions about TRx, NRx, revenue, market share. Use FieldIntelligence for field intelligence, objections, access barriers, and HCP sentiment."*

Snowflake Intelligence: Pre-Loaded Sample Questions

Navigate to: **Snowsight → AI & ML → Snowflake Intelligence → COMMERCIAL_ANALYTICS_AGENT**

1. Which territories are underperforming on Vivitrol TRx vs. target this quarter?
2. Show HCP prescribing trends for Aristada in the Northeast over 12 months
3. Find all psychiatrists in Texas with no rep call in the last 60 days
4. What is the formulary access rate for Vivitrol on commercial plans?
5. Which reps have the highest call-to-prescription conversion rate?
6. Show market share by product across all regions year-to-date
7. Find recent field notes about physician objections or access barriers for Lybalvi
8. Which HCPs in the Midwest have the highest Aristada TRx growth in 6 months?
9. Compare regional performance across all three products YTD with a chart
10. Which payers are creating the most access barriers per rep field reports?

Zero-SQL self-service for the commercial team — click a question, get an answer.

Why Snowflake for Commercial Analytics

- **All data + AI on one platform** — no ETL to external AI services, no API keys, no data copies leaving Snowflake governance
- **No data movement** — Cortex Search indexes live Snowflake tables; Cortex Analyst queries the semantic view directly
- **HIPAA-eligible infrastructure** — Snowflake's BAA covers the full stack including Cortex AI services
- **Enterprise governance built in** — RBAC applies uniformly to the agent, search service, and underlying tables
- **Replaces external vector databases** — no Pinecone, Weaviate, or custom embedding pipelines to maintain or pay for separately
- **Scales with your territory footprint** — add regions, products, or reps by adding rows; no model retraining required
- **Full auditability** — every agent query is logged in `ACCOUNT_USAGE.QUERY_HISTORY`

Next Steps

1. **Run the demo scripts** — six numbered SQL scripts run in under 10 minutes on your Snowflake account
2. **Connect real commercial data** — replace synthetic inserts with IQVIA/Symphony Rx feeds, Veeva CRM call exports, and MCO formulary files
3. **Extend the Semantic View** — add quota attainment %, call frequency targets, reach & frequency metrics, managed care penetration
4. **Deploy to the commercial team** — `GRANT USAGE ON AGENT` to your commercial analyst role; share the Snowflake Intelligence URL
5. **Iterate on sample questions** — after the first week, review top questions and refine the agent instructions and sample question list

Contact your Snowflake account team to get started.

`github.com/sfc-gh-moahmed/pharma-commercial-cortex-demo`